

27) Construct a nonzero 3×3 matrix A and a nonzero vector b such that b is in $\text{Col } A$, but b is not the same as any one of the columns of A .

Solution: By Defn, $\text{Col } A$ represents the set of all linear combinations of the columns of A .

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 3 & 1 \\ -1 & -4 & -2 \end{bmatrix} \quad b = 1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 3 \\ -4 \end{bmatrix} + 1 \begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -7 \end{bmatrix} \in \text{Col } A.$$

28) Construct a nonzero 3×3 matrix A and a vector b such that b is not in $\text{Col } A$.

Solution: By Defn, $\text{Col } A$ represents the set of all linear combinations of the columns of A . Let $A = [a_1 \ a_2 \ a_3]$.

$\forall c_1, c_2, c_3 \in \mathbb{R}, c_1 a_1 + c_2 a_2 + c_3 a_3 \neq b. \Rightarrow Ac = b$ has no solution (By Defn. of Ax)

$$\text{Let, } A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 3 & -3 \\ -1 & -4 & -2 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

By Theorem 3 from Chapter 1, the matrix equation $Ac = b$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ b]$.

$$\begin{bmatrix} 1 & 2 & 4 & b_1 \\ 0 & 3 & -3 & b_2 \\ -1 & -4 & -2 & b_3 \end{bmatrix} \xrightarrow[\text{+ Row 1}]{\text{Row 3 = Row 3}} \begin{bmatrix} 1 & 2 & 4 & b_1 \\ 0 & 3 & -3 & b_2 \\ 0 & -2 & 2 & b_1 + b_3 \end{bmatrix} \xrightarrow[\text{+ } \frac{2}{3} \times \text{Row 2}]{\text{Row 3 = Row 3}} \begin{bmatrix} 1 & 2 & 4 & b_1 \\ 0 & 3 & -3 & b_2 \\ 0 & 0 & 0 & b_1 + b_3 + \frac{2b_2}{3} \end{bmatrix}$$

By Defn, it's easy to see that the final matrix is in echelon form. By Theorem 2 from Chapter 1, $\because Ac = b$ has no solution, $\therefore b_1 + b_3 + \frac{2b_2}{3} \neq 0$

$$b_1 = 1 \quad b_3 = -1 \quad b_2 = 3 \quad b_1 + b_3 + \frac{2b_2}{3} = 2 \neq 0.$$

$$\therefore b = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} \quad \square$$